

Exercises – Inheritance:

1. This question uses inheritance to describe different types of sale in a shop. There are 3 classes to represent the different sale types. The classes should be stored in the same file (Python module) named 'sale_module.py'.
 - 1.1. Define a class named **Sale** that:
 - contains an instance variable that stores the amount of the sale and accessor and mutator methods.
 - contains a method named `__str__` that outputs an English sentence to describe the amount of the sale e.g. "Sale amount: £20.50".
 - includes an appropriate constructor.
 - 1.2. Define a class named **CashSale** that:
 - is derived from **Sale**
 - redefines the `__str__` method to indicate that the payment was made in cash
 - includes an appropriate constructor.
 - 1.3. Define a class named **CreditCardSale** that:
 - is derived from **Sale**
 - contains instance variables for the name on the card, expiration date, and credit card number. The instance variables should have appropriate getters and setters.
 - includes appropriate constructor(s).
 - redefines the `__str__` method to include all credit card information in the printout.
 - 1.4. Define a program named **SaleTester** that:
 - has a main method that creates at least two **CashSale** and two **CreditCardSale** objects with different values and calls `__str__` for each.

These problems are adapted from a course at UCL in the Department of Information Studies, [Programming 1](#).

Further practice problems can be found in the textbook [Think Like a Computer Scientist](#) or the interactive version [How to Think Like a Computer Scientist: Interactive Edition](#)